Qianyu Zhang

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EDUCATION

New York University — **New York, US**
Master of Science in Data Science — **GPA: 3.9**/4.0 — Sep 2024 – Jun 2026
Major Coursework: *Machine Learning, Big Data, Computer Vision, Deep Learning, Natural Language Processing*

University of St Andrews — **St Andrews, UK**
Bachelor of Science (Honours) in Mathematics and Physics — **GPA: 3.8**/4.0 (Dean's List) — Sep 2020 – Jun 2024

EXPERIENCES

Jindong (JD.com) | Fortune Global 500 Company, #44 in 2025 — **Beijing, China**
Data Science Intern — May 2025 – Aug 2025

- Owned and productionized a **user growth ML model** targeting first-time purchase conversion uplift, using XGBoost(AUC = 0.73) to score a **24M**-user daily eligible population; led problem framing and end-to-end execution of experiment-driven strategy across modeling, deployment, and A/B testing, driving **+51%** in experiment and **+45.1%** after full rollout on target channels
- Built a financial LLM **benchmark** by integrating *FinanceIQ*, *FinEval*, and *BizFinBench*; evaluated multiple open-source models across 6 key dimensions to support model selection, routing strategies and quantify performance gains from RAG integration
- Shipped a B2B insurance-agent tool using OCR + DeepSeek to parse policy illustrations into structured features, compute break-even/IRR curves, and visualize cross-product comparisons.
- Delivered **7** BI dashboards and **11** ad-hoc data analyses using **Spark SQL** and Presto

NYU Stern School of Business — **New York, US**
Data Science Research Assistant — Jan 2025 – Jul 2025

- Developed a CrewAI-based **multi-agent system** to automate the academic survey workflow; primary contributor to agent implementation in a *Journal of Marketing Research* under-review manuscript (2025)
- Implemented a Python **web scraping** pipeline using Selenium to collect and preprocess 2,000+ LinkedIn/Instagram posts
- Mined content insights using LDA to identify 5 core thematic clusters and extracted key features such as keyword density and sentiment scores; combined k-means image clustering to generate optimization strategies for NYU Stern's social media posts
- Built an engagement prediction model (XGBoost, $R^2 = 0.6$) using TF-IDF features with PCA compression combined with structured features; deployed as an offline scorer to evaluate LLM-generated posts

Tenbagger Capital — **Hangzhou, China**
Quantitative Research Intern — Jul 2023 – Sep 2023

- Engineered 35 Barra style factors from **1.2M+** trading-day time series and financial-statement records; Implemented a **Barra CNE6** multi-factor performance-attribution framework to attribute excessive return to industry and style factors
- Developed Barra-based risk monitoring with AI industry attribution and alert thresholds to inform portfolio risk management

PROJECTS

Self-Supervised Visual Representation Learning Based on iBOT — Dec 2025

- Reproduced and compared iBOT and DINO under random initialization and <100M backbone constraints; pretrained on 500K unlabeled images and systematically evaluated representation quality using frozen encoder features
- Improved downstream performance via data expansion & linear probing; achieved 33.37% (CUB-200), 73.65% (Mini-ImageNet)

Dataset Distillation for NLP Tasks — Oct 2025 – Dec 2025

- Adapted RaT-BPTT + Boost-DD to NLP by distilling synthetic [CLS] embeddings in continuous embedding space with a frozen sentence encoder; built a PyTorch bilevel-optimization pipeline with warm-start IPC growth on NYU HPC
- Trained lightweight students (MLP, Mini Transformer) on distilled embeddings; evaluated on MRPC/AG News, reaching 72.8% MRPC and 89.6% AG News, with up to +14.4pp over size-matched random real-data baselines

Agentic AI for Automating People Discovery & Outreach | Routed AI Inc. — Sep 2025 – Dec 2025

- Built a LangChain-based agentic workflow to automate people discovery and personalized outreach for recruiters and job seekers; generated channel-ready messages leveraging a 500M+ career-path database and prompt engineering

TECHNICAL PROFICIENCIES

Programming: Python (PyTorch, NumPy, pandas, scikit-learn), SQL, Spark / Hive
Machine Learning: XGBoost, LightGBM, Random Forest; Transformers (MoE, SFT, GRPO, CoT, KV cache, MLA), LangChain, CrewAI
Data Science: A/B Testing, Change Attribution, PSM | **Languages:** English (Fluent; GRE 330, IELTS 7.5), Mandarin (Native)